CLOUD, MICROSERVICES, & APPLICATIONS

MIDTERM EXAMINATION

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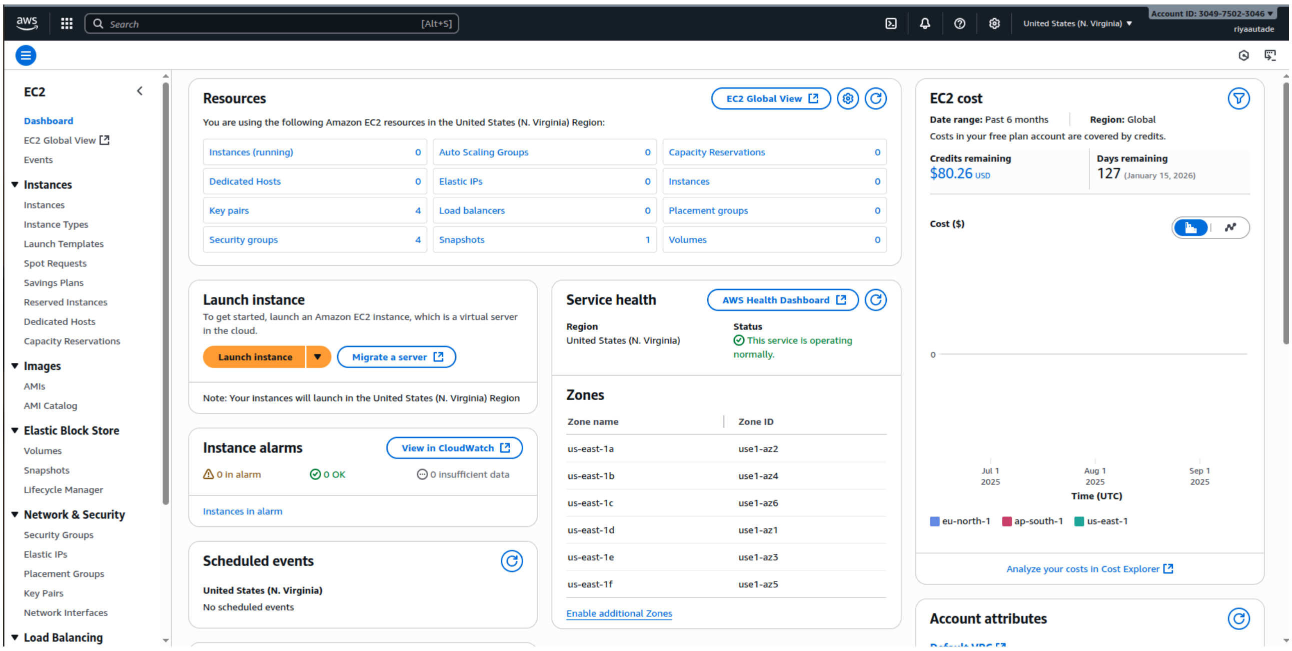
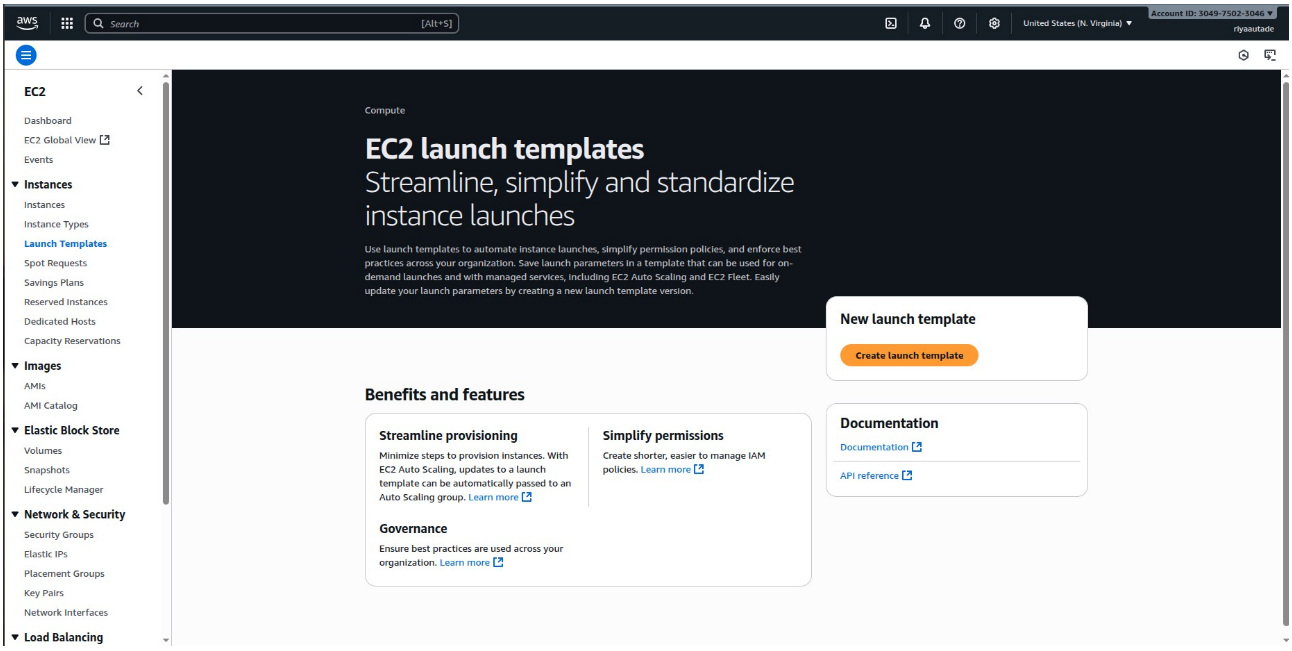
Question:

Implement EC2 Auto Scaling with a Launch Template. Create a Key Pair and a Security Group allowing HTTP (port 80) and SSH (port 22) access. Create a Launch Template using Amazon Linux 2 with a user-data script to install Apache on startup. Create an Auto Scaling Group attached to the default VPC with a minimum of one instance and a maximum of three instances. Configure scaling policies to scale out when CPU utilization exceeds 50% and scale in when it drops below 20%. Test the scaling behavior by generating CPU load and verify Apache access. Document all steps with screenshots.

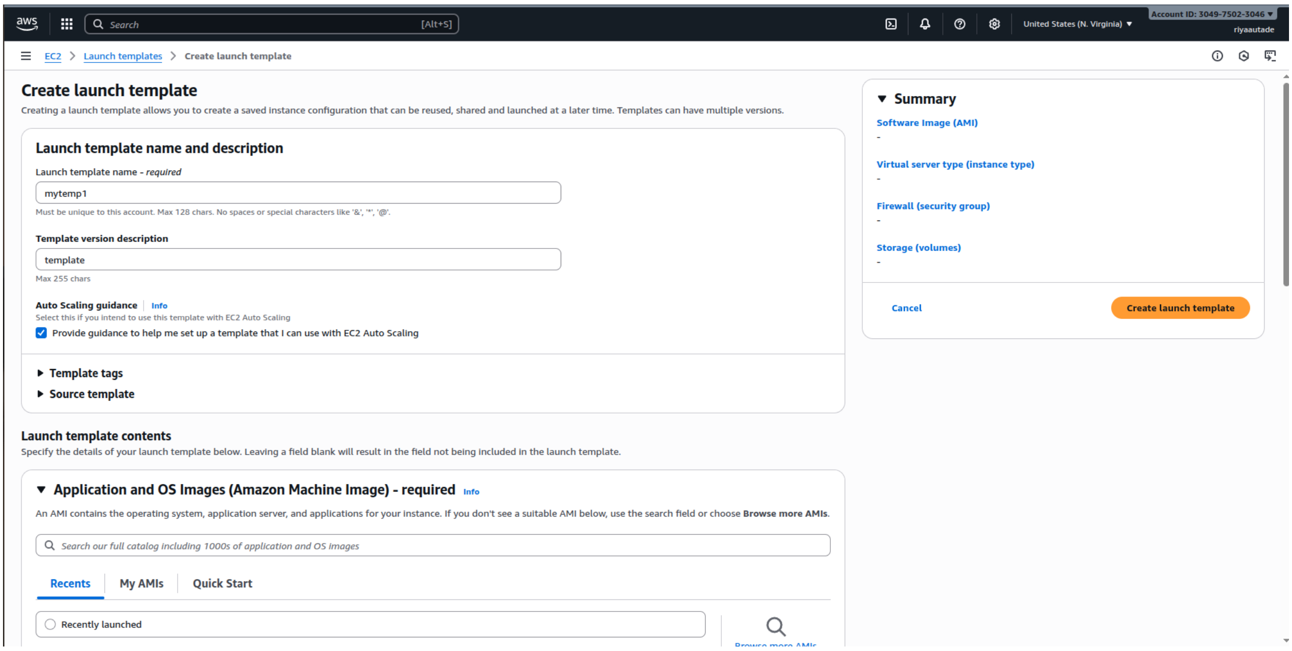
1. Go to EC2 Dashboard and click on Launch Templates:



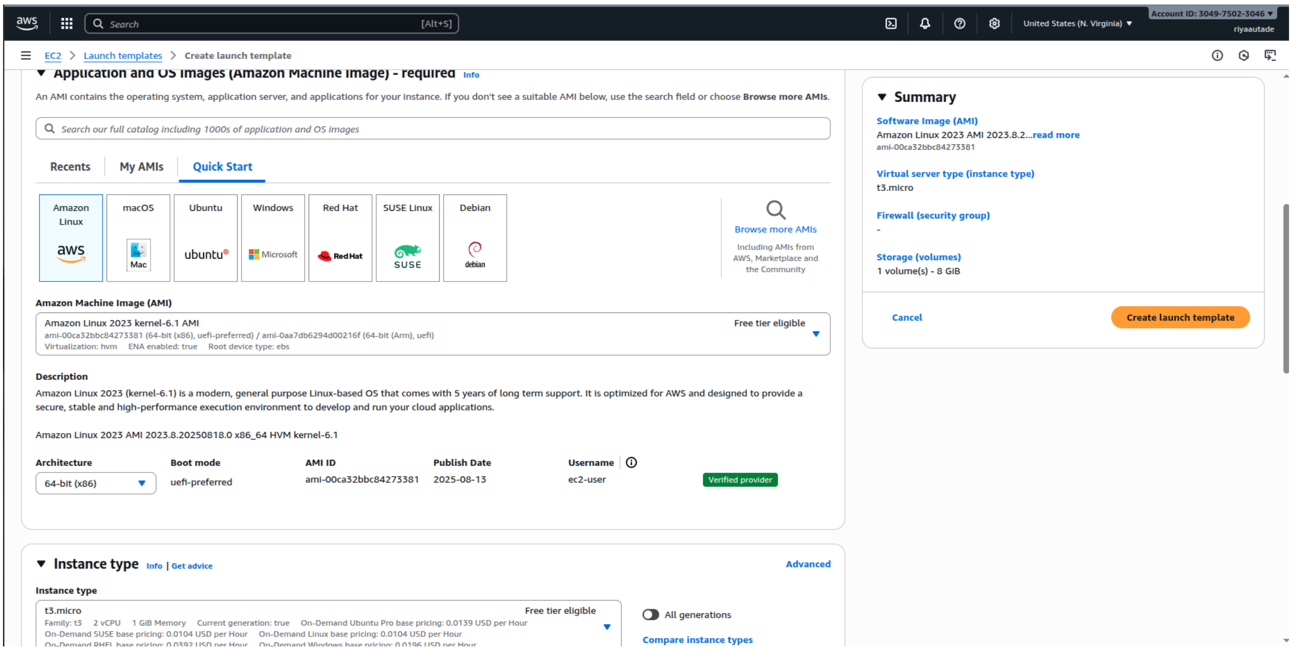
2. Click on Create Launch template



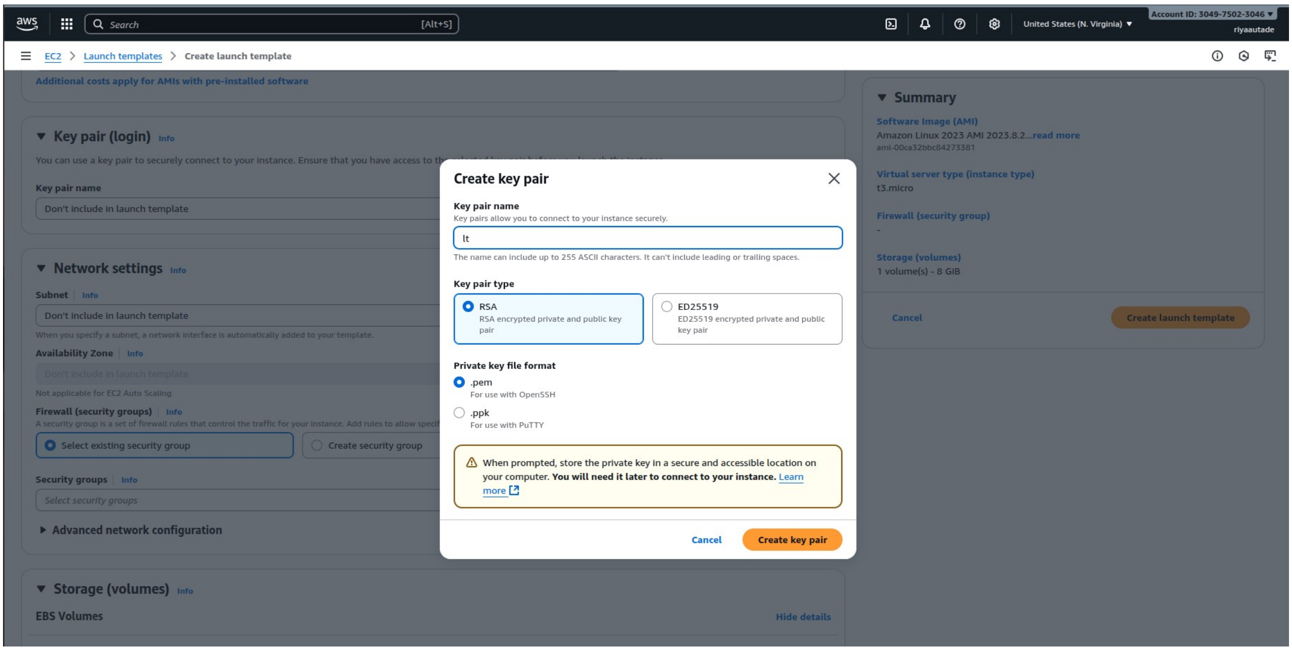
3. Name the template, give a description and choose “Provide Guidance” option.



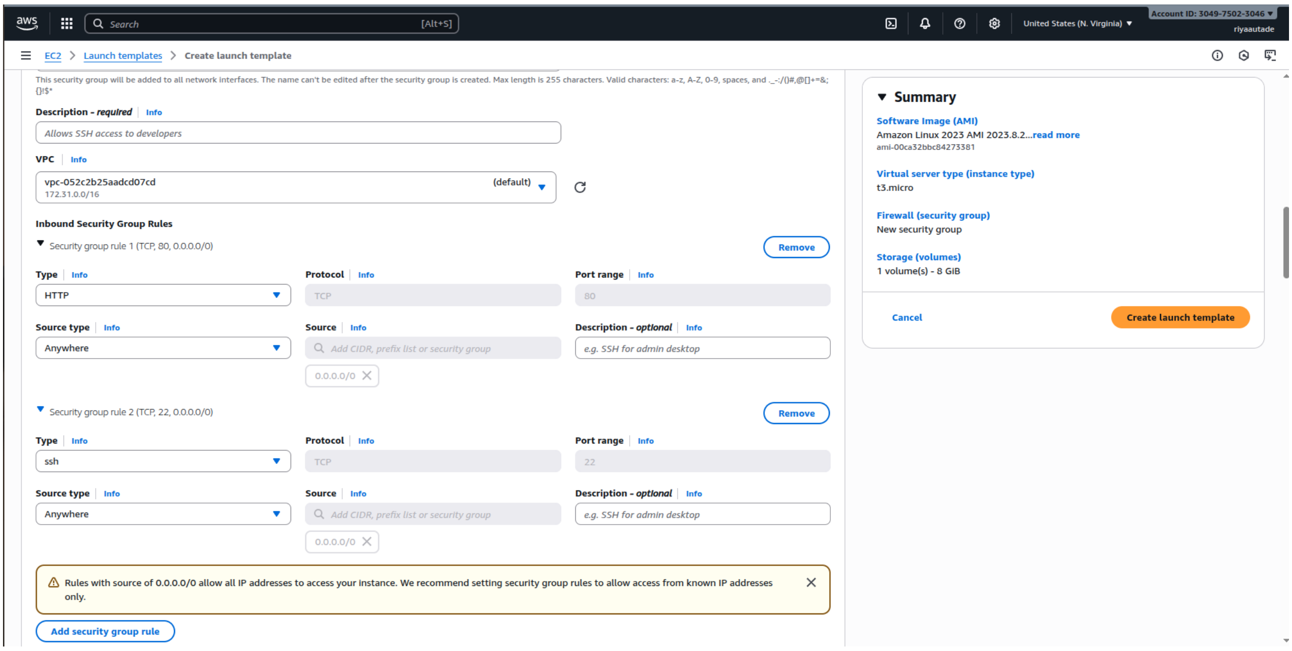
4. Go to Quick Start→ Amazon Linux under AMI. Then choose t3.micro instance



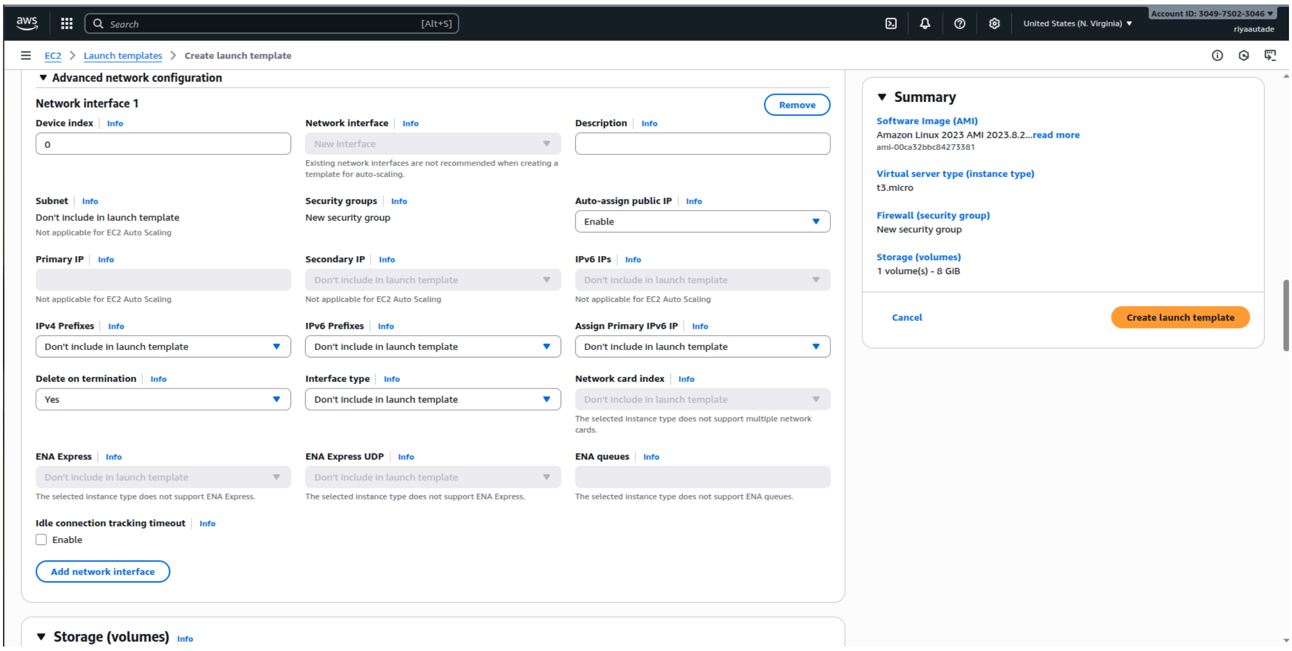
5. Create a key pair, name it, choose type RSA and .pem



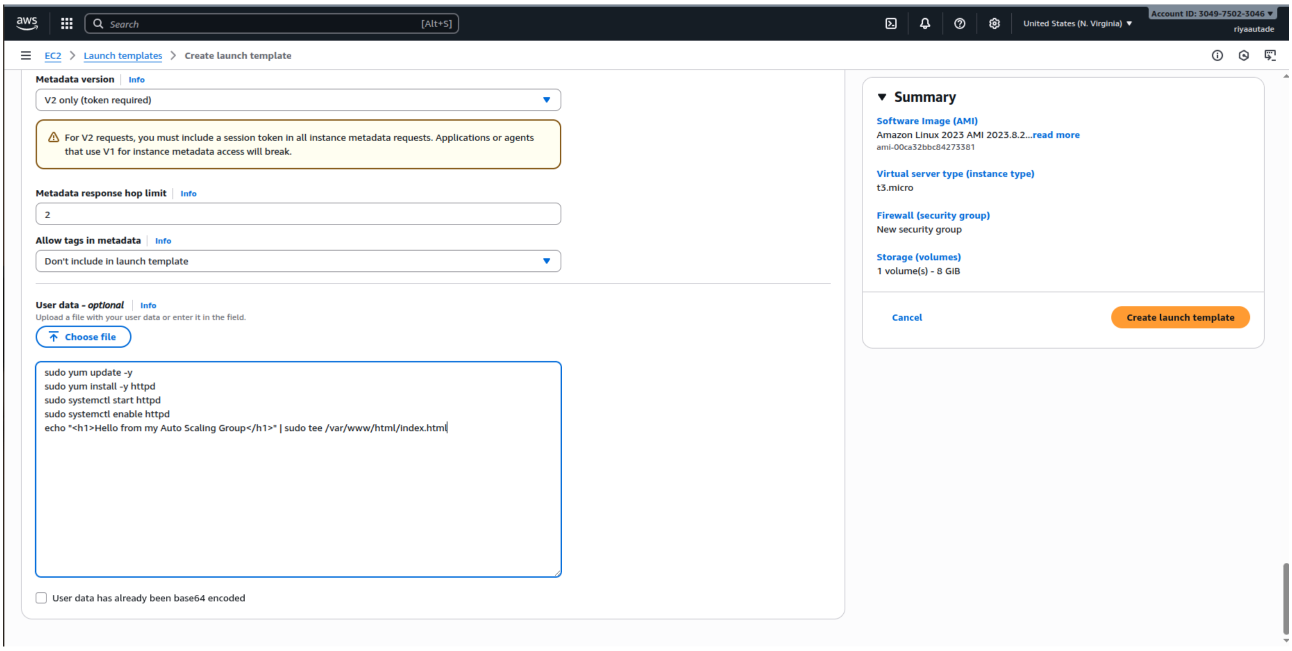
6. Make a new security group with ports 80 and 22 allowed



7. Enable auto assign public IP, and make delete on termination 'Yes'

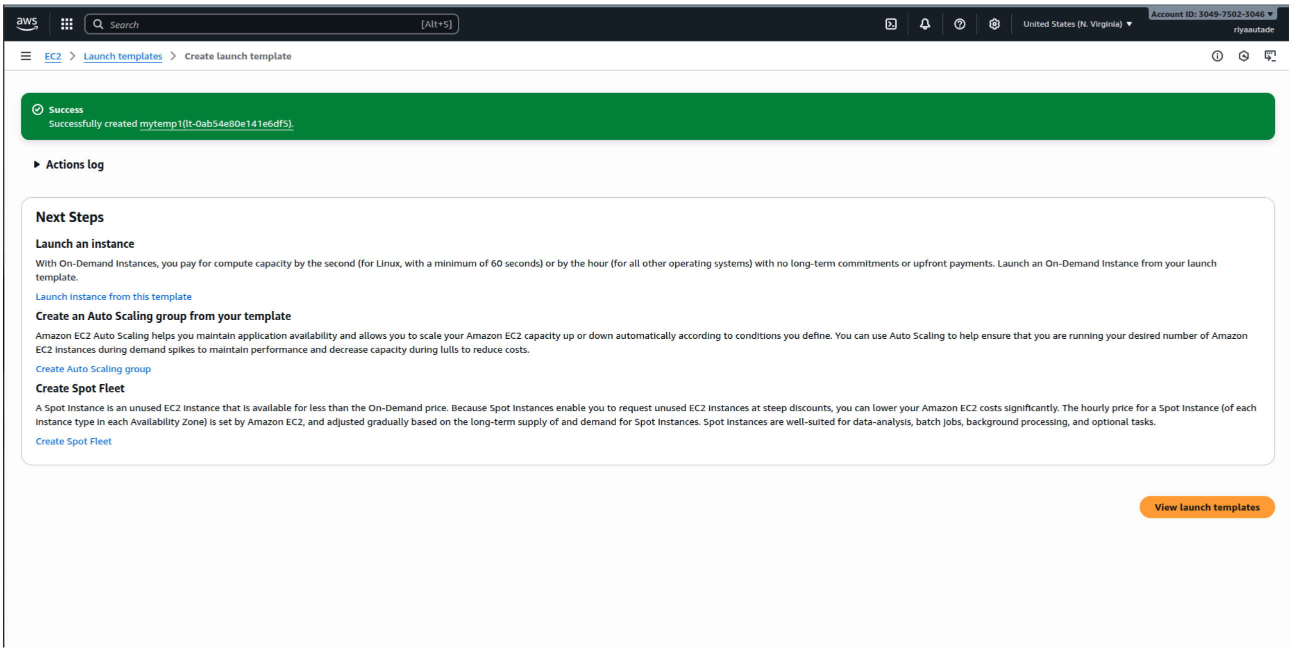


8. Add the user script for Apache

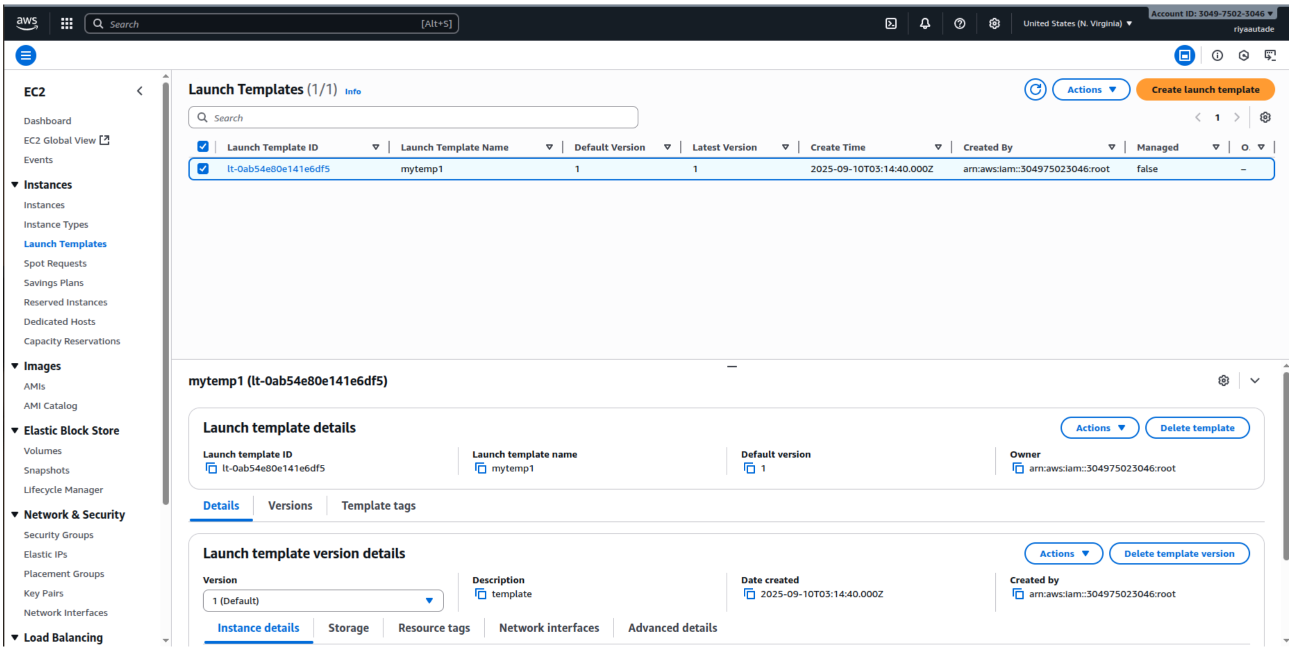


sudo yum update -y
sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
echo "<h1>Hello from my Auto Scaling Group</h1>" | sudo tee /var/www/html/index.html

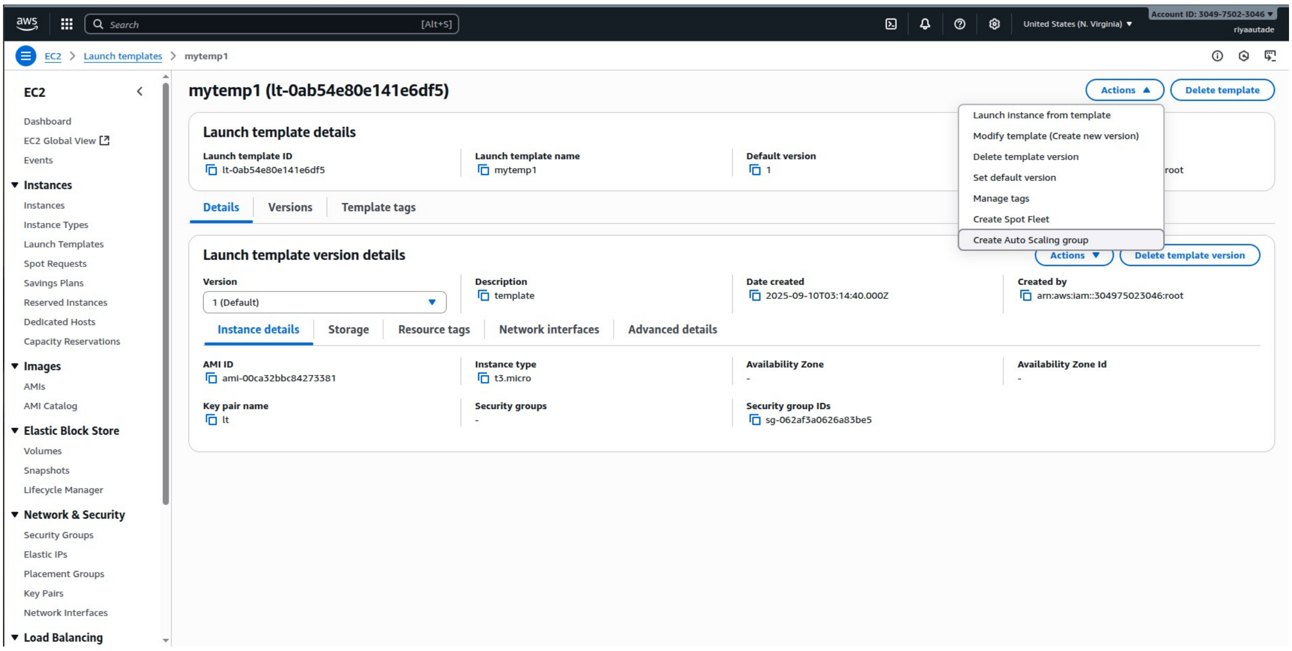
9. Click on create template.



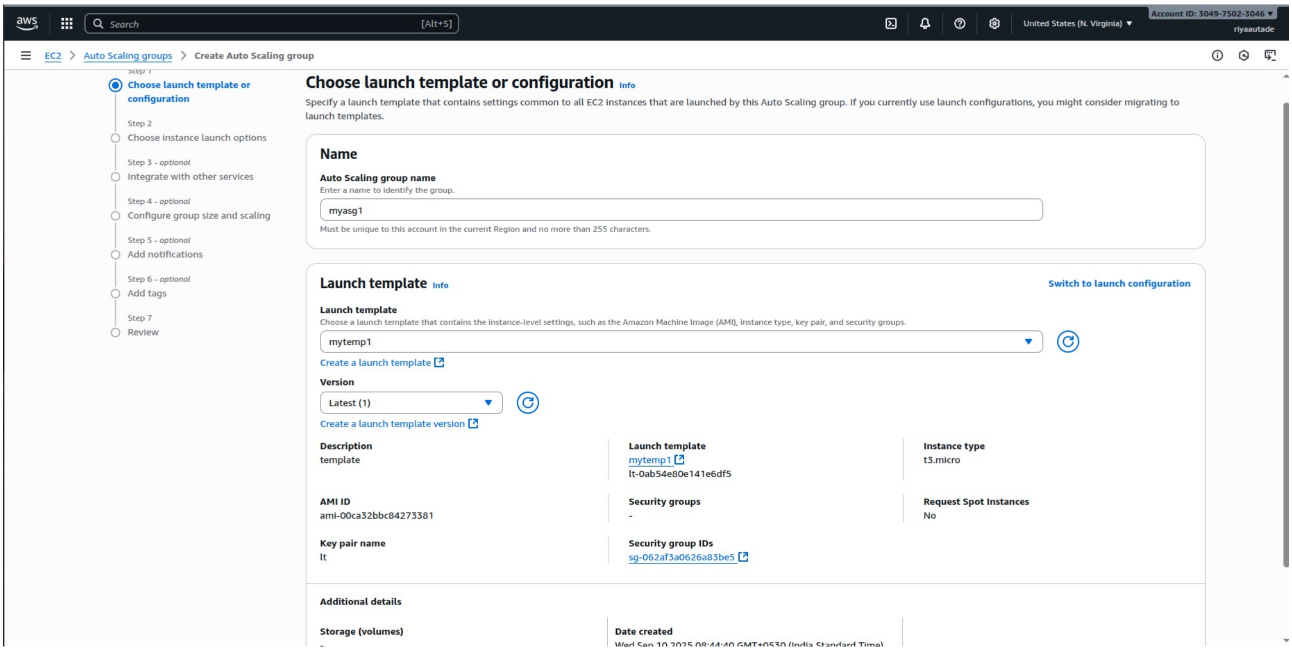
10. view launch template and click on it



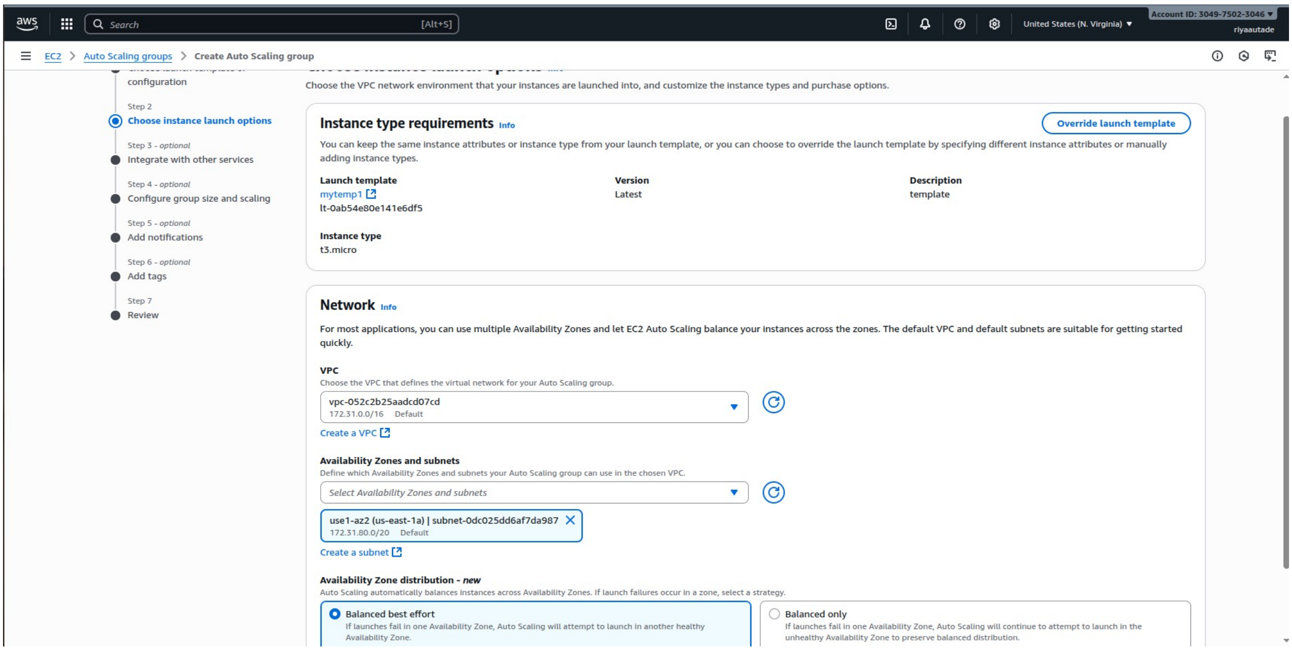
11. Go to Actions → create Auto scaling group



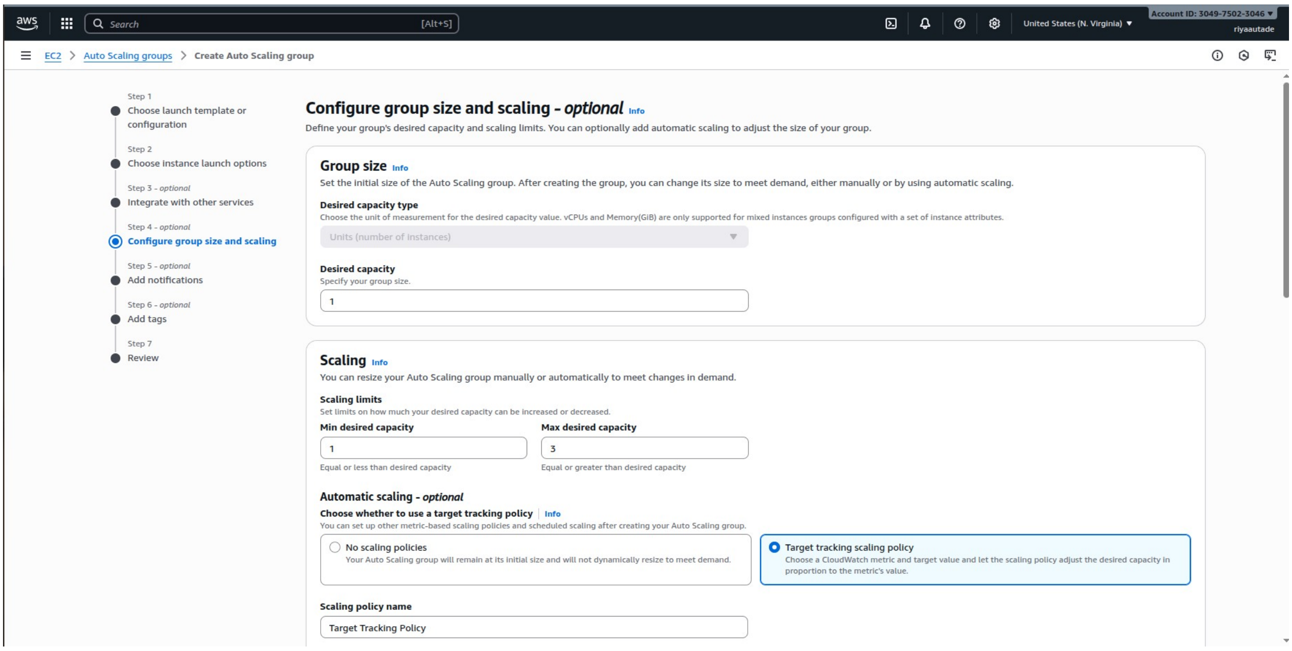
12. name the template ad choose the latest version



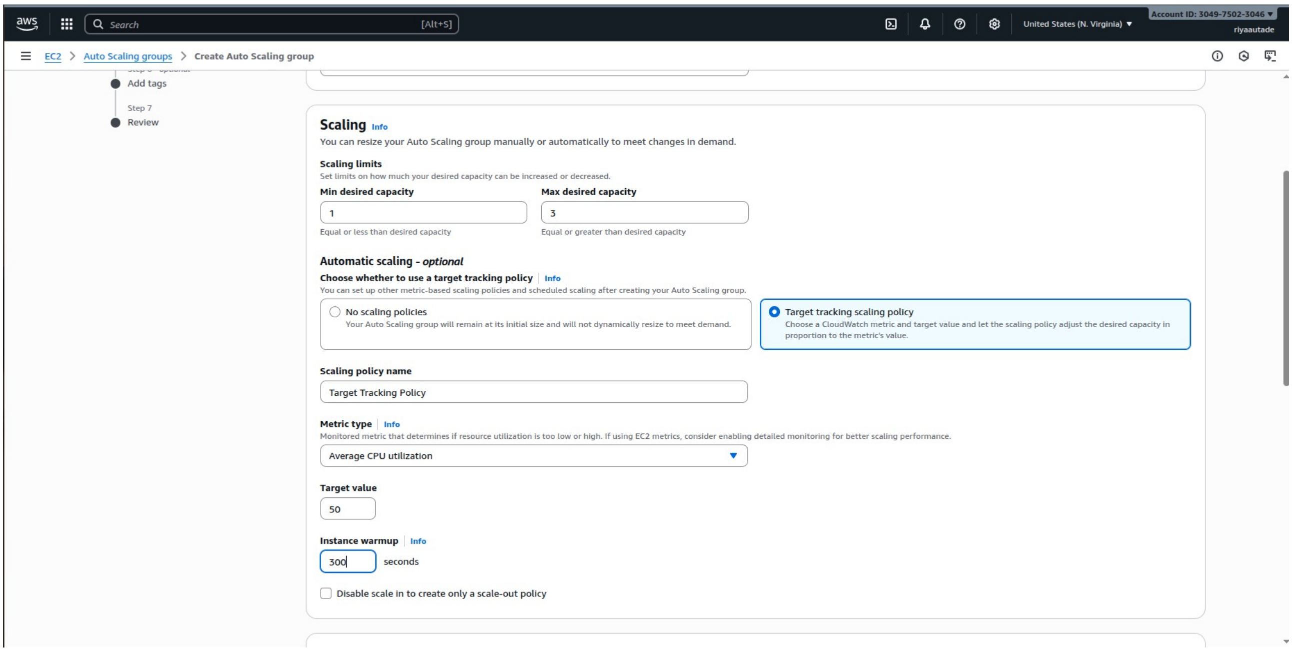
13. Add an availability zone like us-ease-1a and click on Next



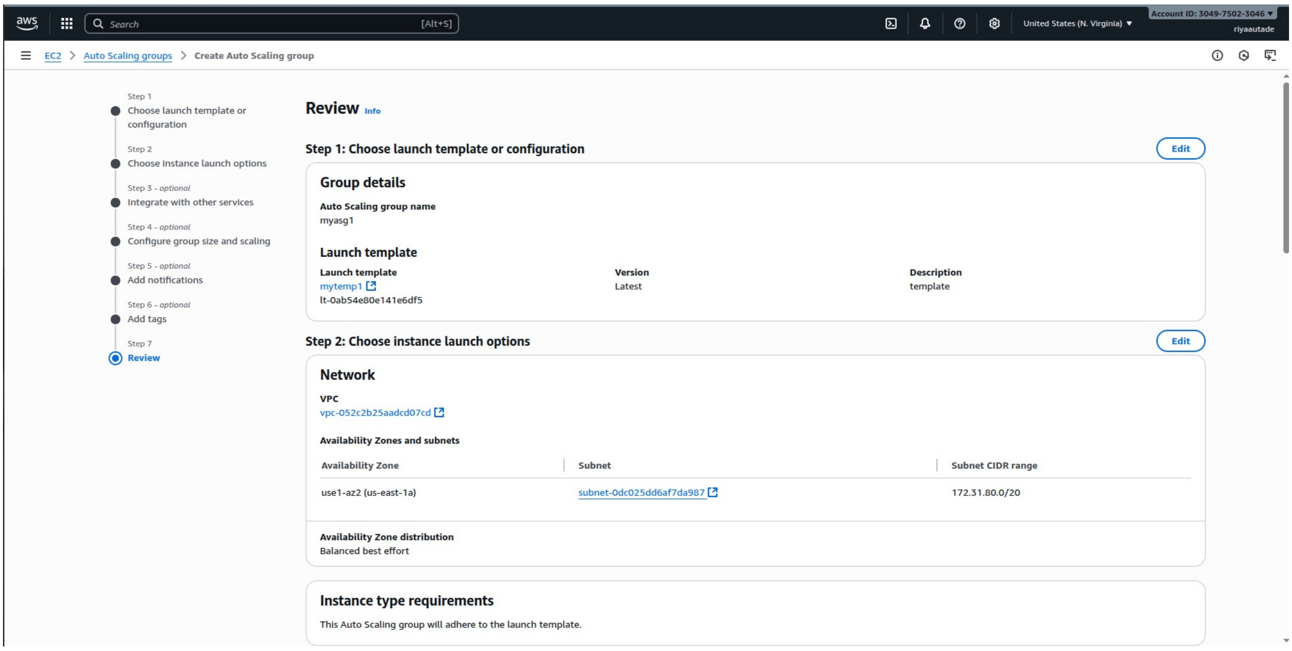
14. Configure group size and scaling as per the question:
Desired Capacity: 1
Minimum Desired Capacity: 1
Maximum Desired Capacity: 3



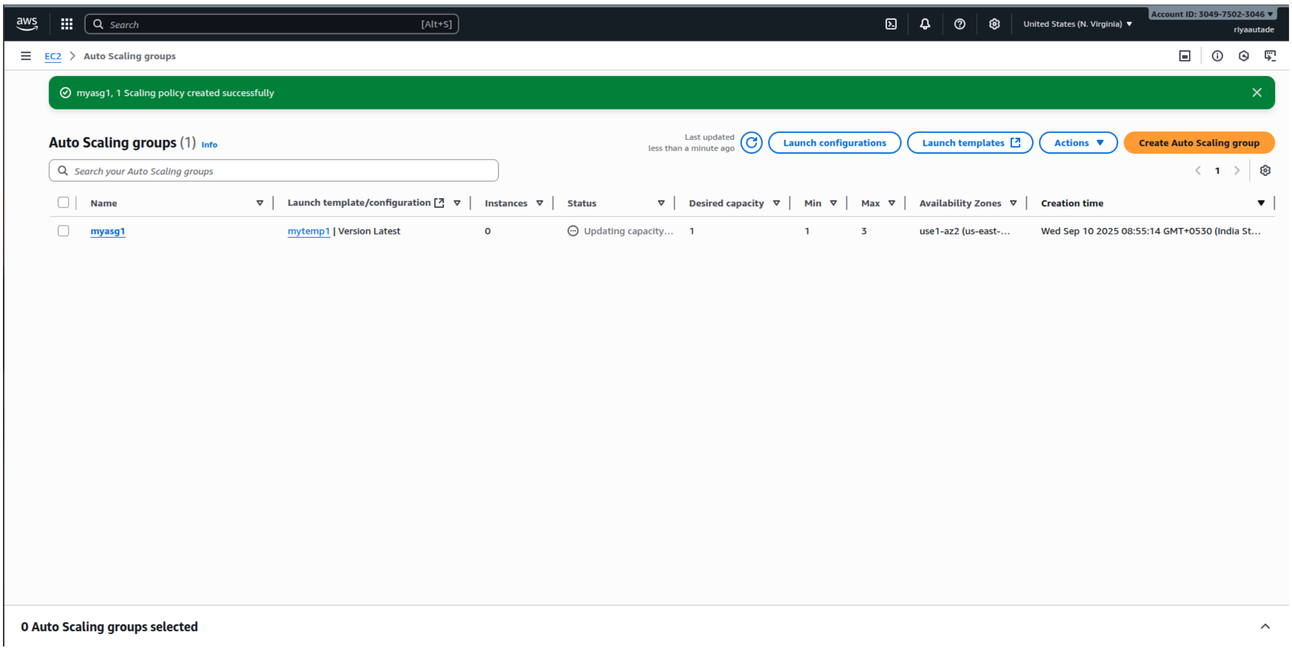
15. Click on Target Tracking Scaling Policy, name it, and click on Average CPU utilization with target value as 50 and instance warnings seconds as 300
(as per the question requirements)



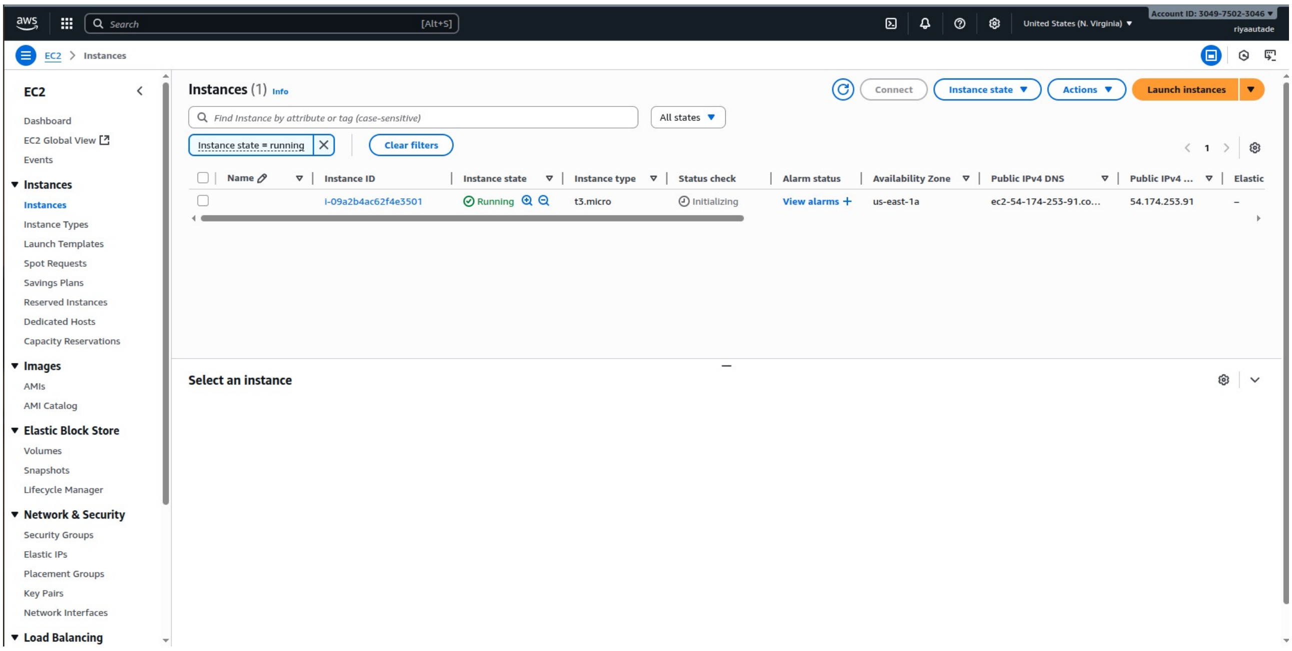
16. Click Next until you reach Review. Then click on create auto scale group



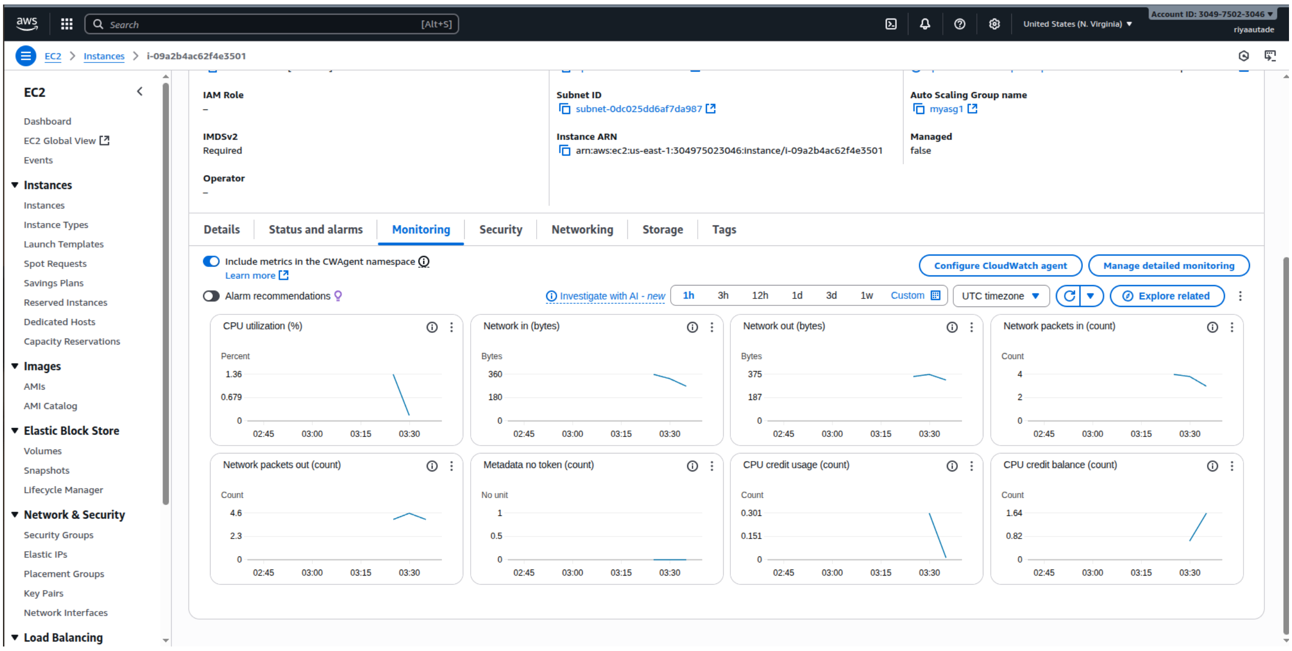
17. Auto Scaling group is successfully created



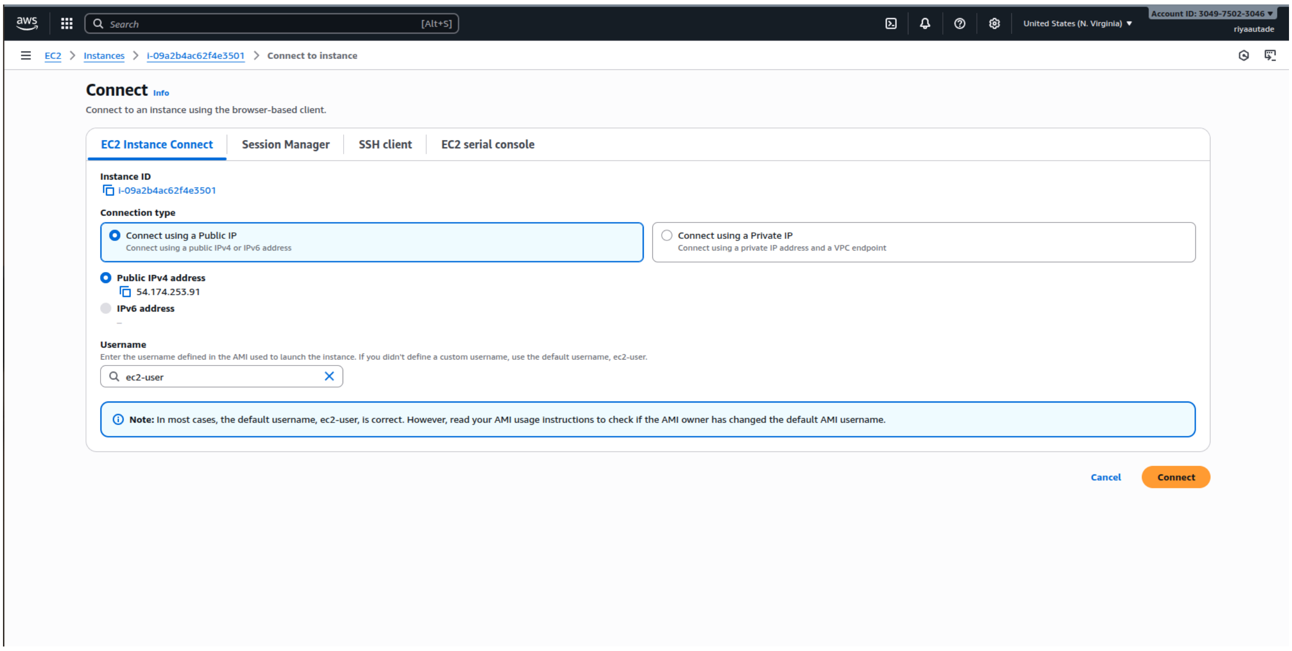
18. EC2 instance for the autoscaling group is visible



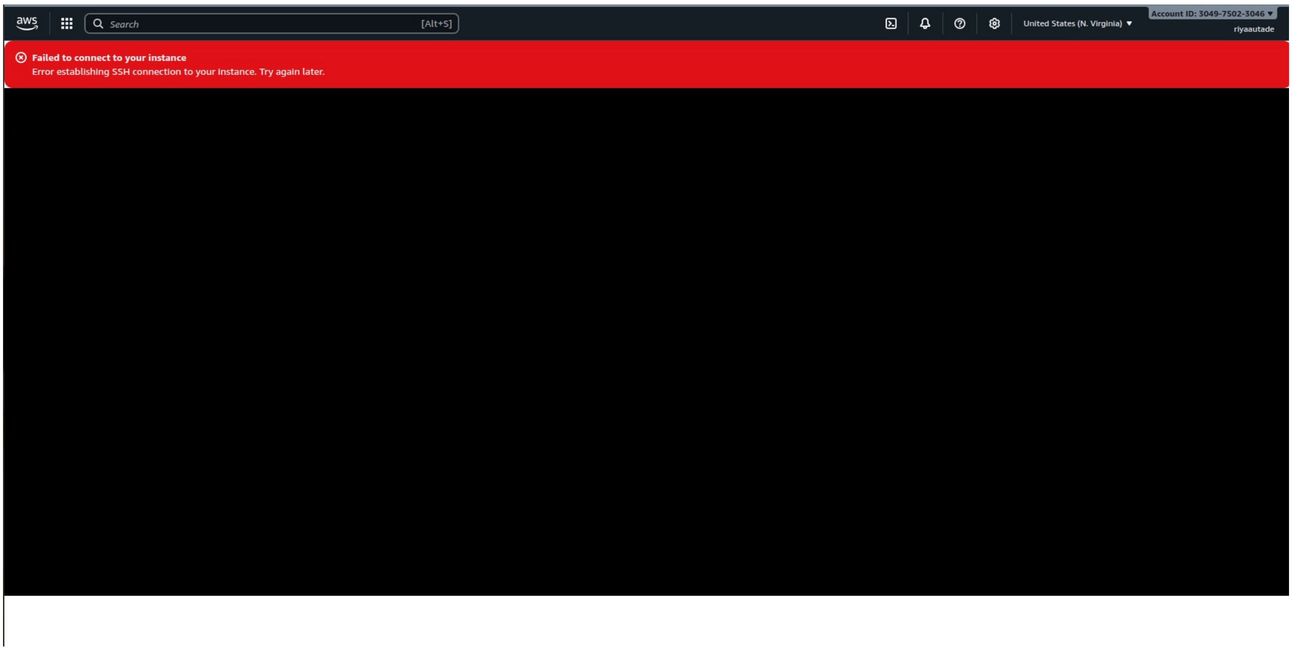
19. Monitoring EC2 results:



20. Connect to SSH



21. SSH Error on the system, therefore cannot verify:



21. Similar issue with public IP for Apache even though the Instance creation and security groups have been properly made and verified: